

Response to Reviewers

Residual-Stream Geometry of Single-Cell Foundation Models
Carries Incremental Gene-Regulatory Signal Across Tissues

Ihor Kendiukhov

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We thank the editor and both reviewers for a constructive and detailed assessment. Every substantive point has been addressed with **new experiments** rather than text changes alone; claim language was then recalibrated in light of the new results. All new analyses are reproducible from the analysis code added to the public repository. Reviewer comments are shown *in italics*; our response and the resulting manuscript changes follow each.

Editor

Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained.

We have (i) added absolute AUROC and AUPRC to every results table, including the cell-type stratification table; (ii) added a leakage-resistant grouped cross-validation analysis that honestly bounds the signal—it collapses under leave-both-out cross-validation; (iii) frozen a confirmatory pipeline applied once to held-out domains; (iv) recalibrated the title, abstract, contributions, Discussion and Conclusions to “incremental signal for retrospective edge prioritization”; and (v) expanded the Limitations from five to seven points. Specifics appear under the reviewer responses below.

Reviewer 1

1. Calibrate core claims more conservatively.

We recalibrated throughout. The **title** now reads “Carries Incremental Gene-Regulatory Signal” (was “Encodes Gene Regulatory Structure”). The abstract, the five-point contribution list, Discussion and Conclusions now use “incremental regulatory-relevant signal for retrospective edge prioritization” and state explicitly that the signal is statistical enrichment, not a stand-alone classifier. The recalibration is grounded in the new grouped-CV results (point 3), not wording alone.

2. Make exploratory vs. confirmatory analyses explicit.

New experiment. A new Methods subsection states that representation choices were selected on kidney and immune. We then **froze** a single pre-specified pipeline (centered cosine, bundle L0–11, stacking; no per-domain tuning) and applied it once to the held-out confirmatory domains. Result: lung Δ AUROC $+0.032 \pm 0.006$, external lung $+0.027 \pm 0.005$ (all three seeds with bootstrap CIs above zero) under edge CV; $+0.0016$ and $+0.0009$ under leave-both-out. Reported in a new Results subsection and table.

3. Strengthen leakage/generalization stress tests.

New experiments. We re-evaluated under leave-TF-out, leave-target-out and leave-both-out grouped CV (new Methods subsection; new Table “Geometric signal under grouped, leakage-resistant cross-validation”). The signal is fully robust to leave-TF-out and leave-target-out (delta comparable to, or larger than, edge CV) but **collapses to near-zero under leave-both-out** (0 to +0.003); the confound baseline also drops to AUROC ≈ 0.50 there. This is now reported as the honest ceiling of out-of-entity generalization, in Results, Discussion and as Limitation 3. We additionally extended the stress test to the compact stacking model: its advantage over the best single branch is preserved under edge-level CV but, like the underlying signal, shrinks under leave-both-out.

4. Expand and clarify negative-edge robustness.

New experiment. A new Methods subsection states the exact protocol (single rejection-sampled draw for the primary set). We additionally generated **degree-matched** and **expression-matched** harder negatives, each repeated for eight independent draws. The signal is stable across draws ($SD \leq 0.042$) and does not weaken under harder negatives—in the lung domains degree-matched negatives *increase* Δ AUROC. New Table “Robustness to negative-edge construction”.

5. Temper the scGPT vs. Geneformer comparison, or make representations comparable.

We chose to **make the extraction comparable** (option a). *New experiment.* We extracted Geneformer per-layer residual-stream activations from real forward passes over single cells (18-layer model, official rank-value tokenization) and applied the identical geometric pipeline to both models. Under matched extraction the apparent Geneformer advantage largely disappears (scGPT ahead in kidney; remaining gaps +0.009–0.023). New Results subsection and Table “Cross-model comparison under matched per-layer extraction”; the original table’s caption now flags its asymmetry; architectural-superiority language has been removed.

6. Improve reporting of absolute performance, uncertainty, multiplicity.

New experiments. Absolute AUROC and AUPRC are now reported in *every* results table—including the cell-type stratification table—and in a full per-layer supplementary panel. A new Methods subsection “Uncertainty decomposition and multiplicity” states the bootstrap unit (edges, or gene clusters under grouped CV), decomposes variance into within-run and across-seed components (across-seed dominates, 64–83%), and applies Benjamini–Hochberg correction across the 148-comparison grid (42 survive $q < 0.05$, including all headline results).

7. Add operational detail for reproducibility.

A new Methods paragraph “Implementation detail” specifies the logistic-regression hyperparameters, in-fold standardization, randomized-SVD PCA configuration, cell-sampling policy, coverage filters and gene/cell aggregation. Every new analysis has a dedicated Methods subsection, and the repository’s README lists reproduction commands for all new analyses.

8. Keep biological interpretation clearly labeled as interpretation.

The CD8⁺ T-cell paragraph now frames the account explicitly as “a biologically plausible hypothesis ... offered as interpretation rather than as an established mechanism” and lists alternative explanations (cell-type abundance, library complexity, annotation density) that are not excluded.

Reviewer 2

1. Report actual AUROC, not just delta.

Done. Every results table now reports baseline-only and geometry-augmented absolute AUROC. Baseline AUROC ranges 0.50–0.65; the geometry-augmented AUROC (0.60–0.69) is always above both 0.5 and the baseline, so the deltas are genuine improvements over a meaningful starting point.

2. Actual AUROC for usability; envision how to sift the signal.

New experiment. A new Results subsection reports top- k retrieval: the geometry-augmented model enriches true edges $1.6\text{--}2.1\times$ over prevalence among the top 50 candidates, and we report a calibrated operating point (precision/recall). We state plainly that the modest absolute AUROC means the signal is an evidence channel for re-ranking, not a stand-alone caller, and the Discussion gives a concrete pipeline (symmetric score for edge presence, asymmetric score for orientation, ensembling with GENIE3).

3. Apply the refined methodology to the earlier results.

Done. The refined pipeline (centered cosine, PCA, bundle) is now applied uniformly to kidney and immune as well; it improves those domains too (kidney $\Delta\text{AUROC } +0.076 \rightarrow +0.122$; immune $+0.024 \rightarrow +0.042$). A new unified-method table replaces the old two-panel table; the Results state that the refinement is a uniform improvement, not a lung-specific rescue.

4. AUROC can be misleading under class imbalance—report AUPRC.

Done. AUPRC and ΔAUPRC are reported in all results tables and the supplementary panel, and precision/recall@ k in the new retrieval analysis. Prevalence is reported per domain so AUPRC can be read against its baseline.

5. Ensemble with existing GRN inference methods—solid evidence.

New experiment. A new Results subsection and Table “Geometric signal on top of expression-based GRN inference”: we computed GENIE3-style random-forest importance and Pearson co-expression on each domain and show that adding the geometric signal improves AUROC by $+0.023$ to $+0.085$ under edge CV, all bootstrap CIs excluding zero. The GRN-implications subsection in the Discussion now cites this direct evidence rather than only envisioning it.

6. Can the geometric signal provide directionality? The metrics appear symmetric.

New experiment. The reviewer is correct that cosine, centered cosine, L_2 and dot product are symmetric and cannot encode direction. We added a directionality analysis using **asymmetric** per-layer features (signed norm and mean-coordinate differences): under pair-grouped CV they predict edge orientation at AUROC 0.80–0.90, and a gene-role classifier separates transcription factors from target-only genes at AUROC 0.74–0.78. A new Results subsection “Asymmetric geometric features carry directional information” presents this; the symmetry of the main metrics is now stated explicitly in Methods and Results.

Summary of new manuscript content

- Revised title; recalibrated abstract, contributions, Discussion, Conclusions.

- New Results subsections: leakage-resistant CV; negative robustness, multiplicity and confirmatory rerun; matched cross-model extraction; directionality; complementing expression-based GRN inference.
- New tables: grouped CV; negative-edge robustness; matched cross-model comparison; GRN ensemble. Tables of initial and cross-domain results rebuilt with absolute AUROC/AUPRC; the cell-type table now carries absolute metrics.
- New Methods subsections: grouped CV; negative sampling; uncertainty and multiplicity; exploratory vs. confirmatory; Geneformer residual-stream extraction; directionality; GRN ensemble; plus an implementation-detail paragraph.
- Limitations expanded from five to seven points.
- All new analysis code and reproduction commands added to the public repository.